

# JUNGJOON LEO KIM

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## EMPLOYMENT

### Postdoctoral Fellow

Department of Physics and Astronomy, York University

2025–

Toronto, ON, Canada

## EDUCATION

### PhD, Physics

Queen's University

2021–2025

Kingston, ON, Canada

Thesis: *On the Formation, Evolution, and Observation of Dark Compact Objects*

Advisor: [Joseph Bramante](#)

### MMath, Applied Mathematics

University of Waterloo

2019–2021

Waterloo, ON, Canada

Thesis: *Spectrum of Cuscuton Bounce and Cosmological Parameter Inference Using Dark Sirens*

Advisor: [Ghazal Geshnizjani](#)

### BMath, Mathematical Physics

University of Waterloo

2014–2019

Waterloo, ON, Canada

## SELECT HONOURS & AWARDS

NSERC Canada Graduate Scholarship – Doctoral (CGS D)

2022–2025

Harold M. Cave Graduate Travel Scholarship

2024

McDonald Institute Student Achievement Award for Equity Leadership

2024

Harold M. Cave Graduate Travel Scholarship

2023

University of Waterloo Applied Math Outstanding Teaching Assistant Award

2021

NSERC Undergraduate Student Research Award

2018

NSERC Undergraduate Student Research Award [Declined]

2016

## PUBLICATIONS AND PREPRINTS

<sup>†</sup>Alphabetical ordering of authors following HEP convention.

- [8] N. Blinov & **J. L. Kim**<sup>†</sup>. *Diffraction of gravitational waves by extended dark objects*. Submitted to Phys. Rev. D. 2026. [\[2609.13369\]](#).
- [7] J. Berger, A. Bhoonah, J. Bramante, **J. L. Kim**<sup>†</sup>, N. Song & L. M. Widrow. *Galactic magnetic fields seeded by ultralight dark photons*. *Phys. Rev. D* **113**.(4) (2026), 043013. [\[2511.07508\]](#).
- [6] D. S. Hosseini, A. Dehghani, **J. L. Kim**, A. Krolewski, S. Mukherjee & G. Geshnizjani. *Modeling gravitational wave bias from 3D power spectra of spectroscopic surveys*. *JCAP* **05** (2026), 029. [\[2506.11201\]](#).
- [5] A. Dehghani, **J. L. Kim**, D. S. Hosseini, A. Krolewski, S. Mukherjee & G. Geshnizjani. *The gravitational wave bias parameter from angular power spectra: bridging between galaxies and binary black holes*. *JCAP* **04** (2025), 056. [\[2411.11965\]](#).
- [4] J. Bramante, M. D. Diamond & **J. L. Kim**<sup>†</sup>. *Dimming Starlight with Dark Compact Objects*. *Phys. Rev. Lett.* **134**.(14) (2025). **Editor's Suggestion**, 141001. [\[2409.08322\]](#).
- [3] J. Bramante, C. V. Cappiello, M. D. Diamond, **J. L. Kim**<sup>†</sup>, Q. Liu & A. C. Vincent. *Dissipative dark cosmology: From early matter dominance to delayed compact objects*. *Phys. Rev. D* **110**.(4) (2024), 043041. [\[2405.04575\]](#).

- [2] J. Bramante, M. Diamond & **J. L. Kim**<sup>†</sup>. *The effect of multiple cooling channels on the formation of dark compact objects*. *JCAP* **02** (2024), 002. [[2309.13148](#)].
- [1] **J. L. Kim** & G. Geshnizjani. *Spectrum of Cuscuton Bounce*. *JCAP* **03** (2021), 104. [[2010.06645](#)].

## PRESENTATIONS

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### — COLLOQUIA AND INVITED SEMINARS —

|                                                                                                                                  |          |
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| <b>Invited Seminar</b> , <i>T-HEP Group – University of Toronto</i><br>“Forming and Finding Dark Compact Objects”                | Dec 2025 |
| <b>Invited Seminar</b> , <i>YITP – Stony Brook University</i><br>“Forming and Finding Dark Compact Objects”                      | Apr 2025 |
| <b>Colloquium</b> , <i>Department of Physics and Astronomy – Queen’s University</i><br>“The secret life of dark compact objects” | Nov 2024 |
| <b>Invited Seminar</b> , <i>Particle Theory Group – UC Irvine</i><br>“Forming and Finding Dark Compact Objects”                  | Sep 2024 |
| <b>Invited Seminar</b> , <i>TEPAPP – UCLA</i><br>“Delayed Dark Clumps and Primordial Black Holes”                                | Sep 2024 |

### — CONFERENCE AND WORKSHOP TALKS —

|                                                                                                                                                                                                       |          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| <b>Contributed Parallel Talk</b> , <i>Dark Interactions 2026 – CERN</i><br>“Constraining extended dark objects with gravitational wave lensing”                                                       | Oct 2026 |
| <b>Contributed Plenary Talk</b> , <i>Light Dark World 2026 – Carleton University</i><br>“Gravothermal collapse beyond minimal models of self-interacting dark matter”                                 | Jul 2026 |
| <b>Contributed Parallel Talk</b> , <i>CITA@40 – CITA</i><br>“Constraining dark compact objects with gravitational wave lensing”                                                                       | May 2026 |
| <b>Contributed Parallel Talk</b> , <i>Dark Side of the Universe 2025 – McGill University/Université de Montréal</i><br>“Dimming Starlight with Dark Compact Objects”                                  | Jul 2025 |
| <b>Contributed Plenary Talk</b> , <i>Dark Interactions 2024 – Simon Fraser University</i><br>“Cloudy with a Chance of Dark Matter”                                                                    | Oct 2024 |
| <b>Contributed Parallel Talk</b> , <i>TeVPA 2024 – University of Chicago</i><br>“Cloudy with a Chance of Dark Matter”                                                                                 | Aug 2024 |
| <b>Contributed Parallel Talk</b> , <i>DPF-PHENO 2024 – University of Pittsburgh/Carnegie Mellon University</i><br>“Delayed Dark Clumps and Primordial Black Holes”                                    | May 2024 |
| <b>Contributed Plenary Talk</b> , <i>Dark Matter, First Light – Perimeter Institute for Theoretical Physics</i><br>“The effect of multiple cooling channels on the formation of dark compact objects” | Feb 2024 |
| <b>Contributed Parallel Talk</b> , <i>PHENO 2023 – University of Pittsburgh</i><br>“More Ways to (Be) Cool: Compact Objects from Inelastic Dark Matter”                                               | May 2023 |
| <b>Contributed Parallel Talk</b> , <i>TeVPA 2022 – Queen’s University</i><br>“A Poisson Log-Normal Framework for Cosmological Parameter Inference Using Dark Sirens”                                  | Aug 2022 |
| <b>Contributed Poster and Lightning Talk</b> , <i>The 9th KIAS Workshop on Cosmology and Structure Formation – KIAS</i><br>“Power spectrum for scalar and tensor perturbations in Cuscuton bounce”    | Nov 2020 |

### — INTERNAL TALKS —

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| <b>Talk</b> , <i>PAGE Conference 2026 – York University</i><br>“Constraining dark extended objects with gravitational wave lensing” | Jun 2026 |
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| <b>Talk</b> , <i>Astronomy Club Student Parade – York University</i>                                                         | Feb 2026 |
| “The secret lives of dark matter stars”                                                                                      |          |
| <b>Talk</b> , <i>Astroparticle Group Meeting – Queen’s University</i>                                                        | Oct 2024 |
| “Cloudy with a Chance of Dark Matter”                                                                                        |          |
| <b>Talk</b> , <i>Astroparticle Group Meeting – Queen’s University</i>                                                        | Feb 2024 |
| “The effect of multiple cooling channels on the formation of dark compact objects”                                           |          |
| <b>Talk</b> , <i>Applied Mathematics Graduate Seminar – University of Waterloo</i>                                           | Jun 2020 |
| “Towards scale invariance in Cuscuton bounce”                                                                                |          |
| <b>Talk</b> , <i>Cosmology Group Meeting – Perimeter Institute for Theoretical Physics</i>                                   | Jun 2020 |
| “Towards scale invariance in Cuscuton bounce”                                                                                |          |
| <b>Talk</b> , <i>Math and Stats Undergraduate Research Conference – McGill University</i>                                    | Aug 2019 |
| “Quadrilateral Regge elements”                                                                                               |          |
| <b>Talk (Best Presentation)</b> , <i>Applied Mathematics Undergraduate Research Mini-Conference – University of Waterloo</i> | Aug 2018 |
| “Power spectrum for Cuscuton bounce”                                                                                         |          |

## MENTORING & TEACHING

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### — MENTORING —

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|---------------------------------------------------------------------------------------------------------|-----------|
| <b>PHYS 3090 Research Course</b>                                                                        | 2026      |
| <i>Navigating the Landscape of Self-Interacting Dark Matter Using Galaxy Rotation Curves</i>            |           |
| Mentees: Salah Hafne, Bayan Taqi (York)                                                                 |           |
| <b>EXPLORE VI Research Course</b>                                                                       | 2025–2026 |
| <i>Navigating the Landscape of Self-Interacting Dark Matter Using Galaxy Rotation Curves</i>            |           |
| Mentees: Altan Gunes, Reamonn Leitzig (Goethe); Salah Hafne, Bayan Taqi, Arthana Sri-Renganathan (York) |           |
| <b>Queen’s University Upper-year Undergraduate Mentorship Program</b>                                   | 2023–2025 |
| <i>Graduate school and research assistant application mentorship</i>                                    |           |
| Mentees: Alexander Katrusiak, Francesca Valentini (Queen’s)                                             |           |
| <b>Summer Undergraduate Research Project</b>                                                            | 2020      |
| <i>Cross-correlation of the Gravitational Wave Background with Galaxy Surveys</i>                       |           |
| Mentees: Kieana Fana (Waterloo); Jordan Krywonos, Madison Tindall (Perimeter)                           |           |

### — TEACHING —

|                                                                                |                             |
|--------------------------------------------------------------------------------|-----------------------------|
| <b>Graduate Teaching Assistant</b>                                             | 2021–2025                   |
| <i>Queen’s University</i>                                                      | <i>Kingston, ON, Canada</i> |
| · PHYS 345: Quantum Physics of Atoms, Nuclei and Particles (Winters 2022–2025) |                             |
| · Physics Help Desk (Winter 2024)                                              |                             |
| · APSC 112: Physics II (Winter 2023)                                           |                             |
| · PHYS 316: Methods in Mathematical Physics I (Falls 2021–2022)                |                             |
| · PHYS 344: Introduction to Quantum Mechanics (Fall 2021)                      |                             |
| <b>Graduate Teaching Assistant</b>                                             | 2019–2021                   |
| <i>University of Waterloo</i>                                                  | <i>Waterloo, ON, Canada</i> |
| · MATH 674: Special Relativity for Teachers (Spring 2021)                      |                             |
| · AMATH 373: Quantum Theory 1 (Winter 2021)                                    |                             |
| · MATH 228: Differential Equations for Physics and Chemistry (Winter 2021)     |                             |
| · AMATH 456: Calculus of Variations (Fall 2020)                                |                             |
| · MATH 636: Linear Algebra for Teachers (Spring 2020)                          |                             |

- AMATH 353: Partial Differential Equations 1 (Winter 2020)
- MATH 217: Calculus 3 for Chemical Engineering (Winter 2020)
- MATH 115: Linear Algebra for Engineering (Fall 2019)

**Undergraduate Teaching Assistant**

*University of Waterloo*

2016–2019

*Waterloo, ON, Canada*

- MATH 137: Calculus 1 for Honours Mathematics (Falls 2016, 2018)
- MATH 138: Calculus 2 for Honours Mathematics (Winters 2017, 2019)
- ECE 206: Advanced Calculus 2 for Electrical Engineers (Fall 2017)

**OUTREACH**

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|-----------------------------------------------------------------------------|-----------|
| <b>Public Talk</b> , <i>Observatory Open Night, Queen's University</i>      | 2025      |
| <b>Public Education Specialist</b> , <i>McDonald Institute</i>              | 2022–2024 |
| <b>Summer Camp Counselor</b> , <i>IDEAS Summer Camp, Queen's University</i> | 2023      |
| <b>Volunteer</b> , <i>Science Rendezvous Kingston</i>                       | 2023      |

**SERVICE**

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|----------------------------------------------------------------------------------------------------------------|-----------|
| <b>Referee</b> , <i>JCAP</i>                                                                                   | 2026–     |
| <b>HEP Journal Club Organizer</b> , <i>York University</i>                                                     | 2025–     |
| <b>Founding Member, Mentor, Executive</b> , <i>Queen's Physics Upper-year Undergraduate Mentorship Program</i> | 2023–2025 |
| <b>Colloquium Committee Graduate Rep</b> , <i>Queen's University</i>                                           | 2022–2024 |
| <b>Volunteer</b> , <i>TeVPA 2022 @ Queen's University</i>                                                      | 2022      |